

ShadowTrace AI

5

SIGNAL LAYERS

200+

ANNOTATED POSTS

87%

VALIDATION ACCURACY

3

RISK TIERS

XAI

EXPLAINABLE

OVERVIEW

Cybercrime actors frequently operate multiple aliases across social media platforms to evade detection and impersonate legitimate services. Manually correlating these accounts is slow, inconsistent, and difficult to sustain across active investigations. Unlike conventional keyword or metadata-matching tools, **ShadowTrace AI operates on behavioral patterns** — making it resilient to alias changes and account recycling. The system is a SOCMINT prototype that combines behavioral fingerprinting and Explainable AI to identify coordinated digital identities from publicly available data.

TECHNICAL APPROACH

The system extracts **five behavioral signals** across submitted handles — posting timing patterns, multilingual phrase similarity (Hindi + English fraud templates), hashtag cluster overlap, stylometric writing analysis, and repeated payment link patterns. Similarity scoring uses multilingual sentence-transformer embeddings. A weighted explainable scoring engine produces a **0–100% Coordinated Activity Score**, classified into three actionable tiers: **HIGH** (escalate), **MEDIUM** (monitor), and **LOW** (archive).

| # | SIGNAL                         | METHOD                                             | WEIGHT |
|---|--------------------------------|----------------------------------------------------|--------|
| 1 | Posting Timing Patterns        | Cross-account temporal delta analysis              | 20%    |
| 2 | Multilingual Phrase Similarity | Sentence-transformer embeddings (HI+EN)            | 25%    |
| 3 | Hashtag Cluster Overlap        | Jaccard similarity on tag sets                     | 20%    |
| 4 | Stylometric Analysis           | Punctuation, emoji, sentence structure fingerprint | 20%    |
| 5 | Payment Link Patterns          | Shortened URL hash matching across handles         | 15%    |

INVESTIGATOR OUTPUT

Each analysis generates a **plain-language XAI reasoning summary** explaining which signals triggered the verdict. A timestamped incident report is produced with legal references (IT Act §66C, §66D; IPC §420), relationship graph visualisation, and a recommended action — formatted for direct handoff to a cybercrime unit.

| TIER   | SCORE RANGE | ACTION                                  | INDICATOR |
|--------|-------------|-----------------------------------------|-----------|
| HIGH   | 70 – 100%   | Escalate to cybercrime unit immediately | ●         |
| MEDIUM | 40 – 69%    | Place under active monitoring           | ●         |
| LOW    | 0 – 39%     | Archive; flag for periodic review       | ●         |

SYSTEM INTERFACE — LIVE PROTOTYPE

The interface accepts two social media handles, runs the five-signal behavioral pipeline in real time, and renders a live **Coordinated Activity Score** with full XAI reasoning breakdown and interactive relationship graph. A downloadable timestamped incident report (.txt) is generated on completion. The demo mode runs on a pre-loaded anonymised OSINT dataset of 200 ethically collected public posts — no live API calls are required during demonstration.

SHADOWTRACE AI

EXPLAINABLE BEHAVIORAL SCORING • SOCINT INVESTIGATION SUPPORT

Behavioral engine ready • OSINT dataset loaded • XAI v2.1

5

SIGNAL LAYERS

200+

ANNOTATED POSTS

3

RISK TIERS

XAI

EXPLAINABLE

INVESTIGATION INPUT

CIDCODE 2026 - Pse3

@ crypto\_helpdesk7

@ upi\_refund\_team

RUN INVESTIGATION

CLEAR

Demo mode: Running on pre-loaded anonymised OSINT dataset (200 ethically collected public posts) • IF Act 2006 compliant • CERT-In aligned

BEHAVIORAL RISK SCORE

dataset

91%

COORDINATED ACTIVITY ASSESSMENT

HIGH RISK • COORDINATED

ACTIVITY SCORE

Posting timing overlapdataset89%

Multilingual phrase matchdataset84%

Hashtag cluster overlapdataset87%

Writing style similaritycomputed72%

Payment link patternsdataset95%

BEHAVIORAL RISK SCORE

dataset

91%

COORDINATED ACTIVITY ASSESSMENT

HIGH RISK • COORDINATED

ACTIVITY SCORE

Posting timing overlapdataset89%

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Hashtag cluster overlapdataset87%

Writing style similaritycomputed72%

Payment link patternsdataset95%

DETECTED ACCOUNT CLUSTER

@crypto\_helpdesk7

@upi\_refund\_team

@crypt0\_support

Shared links detected

Jaccard

Leet

Token

Multilingual

XAI v2.1

Char. Similarity

Normalisation

Overlap

Phrase Match

Explainable

RELATIONSHIP GRAPH

NetworkX

behavioral similarity

@crypto\_helpdesk7

@crypt0\_support

@upi\_refund\_team

Shared Links

70% sim

70% sim

Coordinated

Payment

Payment

4 nodes • behavioral similarity graph • XAI v2.1

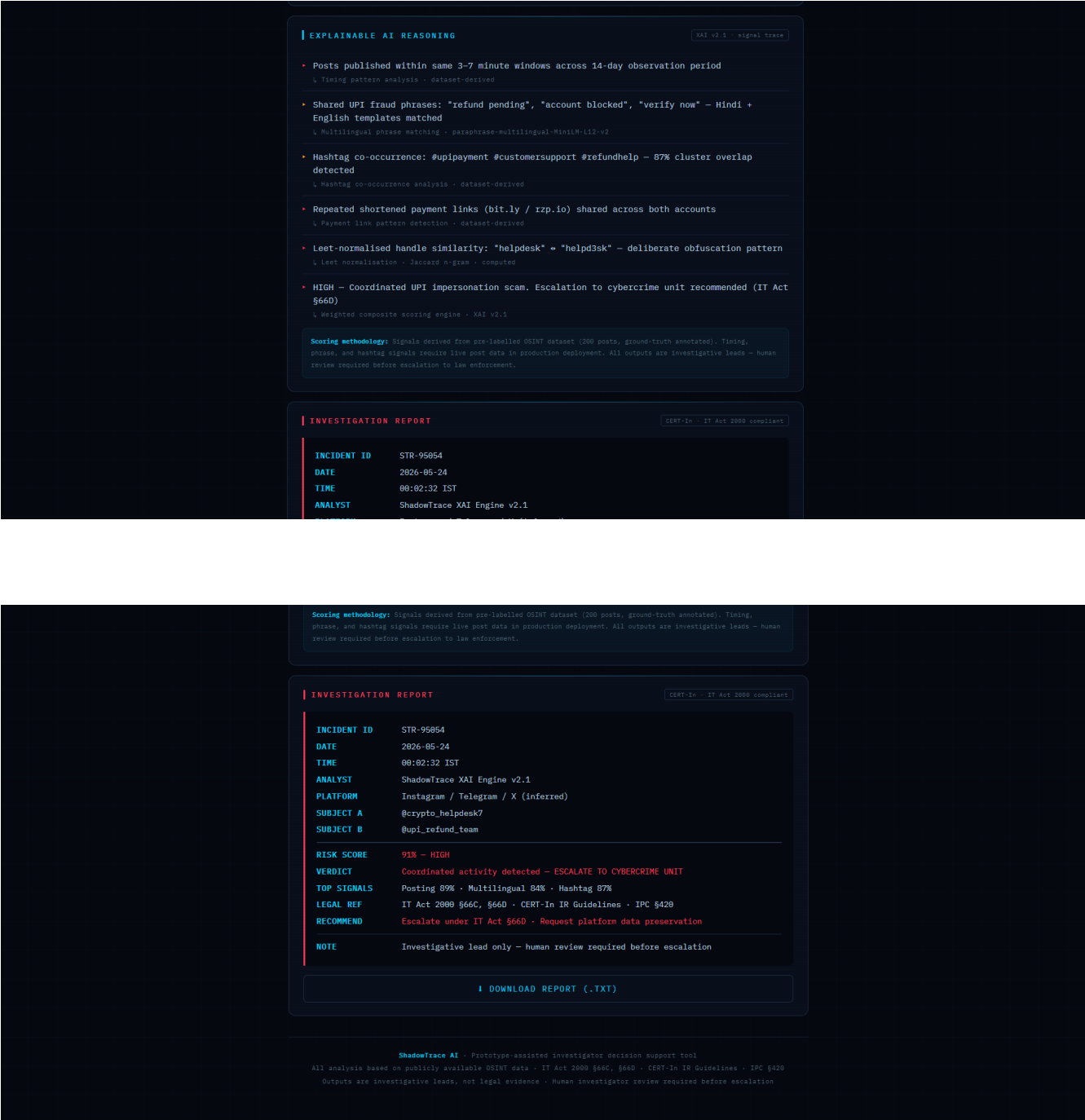


Fig 1. ShadowTrace AI live investigator interface — Investigation Input panel with behavioral engine status indicators, demo dataset confirmation, and legal compliance footer.

EXAMPLE BEHAVIORAL INDICATORS

| SIGNAL         | EXAMPLE INDICATOR                                       | FRAUD VECTOR             |
|----------------|---------------------------------------------------------|--------------------------|
| Phrase Match   | Repeated phrases: 'refund pending', 'verify now'        | Phishing / Impersonation |
| Timing Pattern | Accounts posting within identical 5-minute windows      | Coordinated Botnet       |
| Stylometry     | Consistent punctuation, emoji usage, sentence structure | Same-author fingerprint  |
| Payment Links  | Shared shortened payment URLs detected across handles   | UPI / Crypto fraud       |

|                 |                                                     |                       |
|-----------------|-----------------------------------------------------|-----------------------|
| Hashtag Overlap | Identical hashtag clusters across separate accounts | Amplification network |
|-----------------|-----------------------------------------------------|-----------------------|

COMPLIANCE & VALIDATION

87%

**Validation Accuracy**

System verdicts vs. ground-truth annotations across 200 anonymised, ethically collected public posts across manually labelled account pairs. Behavioral signal extraction and similarity scoring validated through structured test protocol.

All analysis operates exclusively on publicly available OSINT data, aligned with **CERT-In Incident Response Guidelines** and **IT Act 2000**. The scoring pipeline is deterministic, ensuring reproducibility across repeated analyses. A functional end-to-end prototype is complete and available for live demonstration.

|                           |          |                    |               |                    |
|---------------------------|----------|--------------------|---------------|--------------------|
| IT Act 2000<br>§66C, §66D | IPC §420 | CERT-In<br>Aligned | OSINT<br>Only | XAI<br>Explainable |
|---------------------------|----------|--------------------|---------------|--------------------|

DEPLOYMENT & DEMONSTRATION

The prototype runs entirely client-side in a browser with no backend dependency, enabling immediate deployment without infrastructure setup. The system is designed as a deployable investigator-support workflow with interactive analysis, relationship graph visualisation, deterministic behavioral scoring, and downloadable incident report generation. Analysis of a submitted account pair completes in under 3 seconds. The system is intended for demonstration in cybercrime investigation and SOCMINT evaluation environments.

Live Prototype: <https://haresh03072008-cyber.github.io/CIDECODE-Hackathon---2026/>